

Week 5 - Data Modelling

Power BI - The Insight Stack

What data modelling does

Model View: Structure tables, relationships and fields for reporting



Understand your model components

Tables

Two types to know

- ✓ Fact: events and transactions
- ✓ Dimension: filters and groups
- ✓ Date table is essential
- ✓ Keys connect the tables

Relationships

How tables connect

- ✓ One-to-Many preferred
- ✓ Many-to-Many with caution
- ✓ Single filter direction default
- ~ Both direction when needed

Schema Design

How the model is arranged

- ✓ Star schema preferred
- ✓ Simple and fast for reports
- ✓ Snowflake for complex sources
- ~ More joins, more complexity

Key questions before modelling

Which table is the fact table?

Are join keys matching?

What is the granularity?

Is cardinality correct?

Active or inactive relationship?

Does the model need a Date table?

"Data modelling decides how your tables work together."

Next up: Week 6 - DAX

Introduction

When a model has more than one table, appropriate relationships need to exist between those tables. Once the data has been connected, cleaned, shaped, and checked for quality, it needs to be organised into a model. In Power BI, data modelling is the process of structuring tables, relationships, and fields so the report can calculate, filter, and display information correctly. The relationships allow filters applied on one table to propagate to another table, as long as there is a valid relationship path to follow.

In simple terms: **Data modelling decides how your tables work together.** A strong model helps Power BI answer business questions clearly and consistently.

Fact Tables and Dimension Tables

A good Power BI model usually separates data into two main types of tables: fact tables and dimension tables.

Fact Tables: Fact tables store business events, observations, or transactions.

Fact tables usually contain:

- Keys that connect to dimension tables
- Numeric values that can be aggregated
- Transaction or event-level records

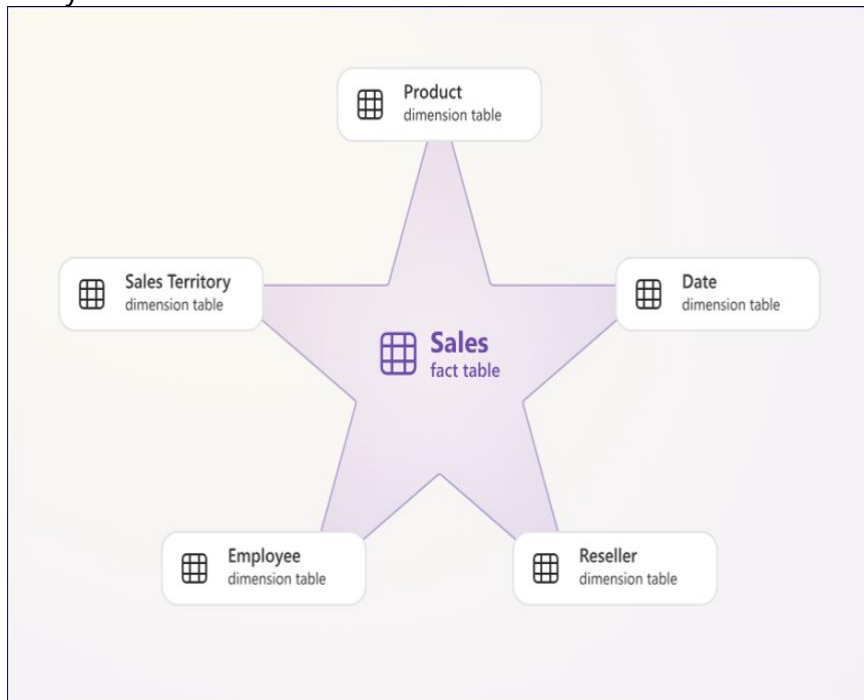
The keys define the level of detail, or **granularity**, of the fact table.

Dimension Tables: Dimension tables describe the business entities used to filter, group, or explain the facts.

A dimension table usually contains:

- A key column that uniquely identifies each row.
- Descriptive columns used for filtering and grouping

The most common dimension table in many models is the **Date table**, because most business analysis involves time.



Relationships in Power BI

Relationships connect tables inside the model. Each relationship has:

- One “from” columnOne “to” column
- Matching values between those columns
- Compatible data types

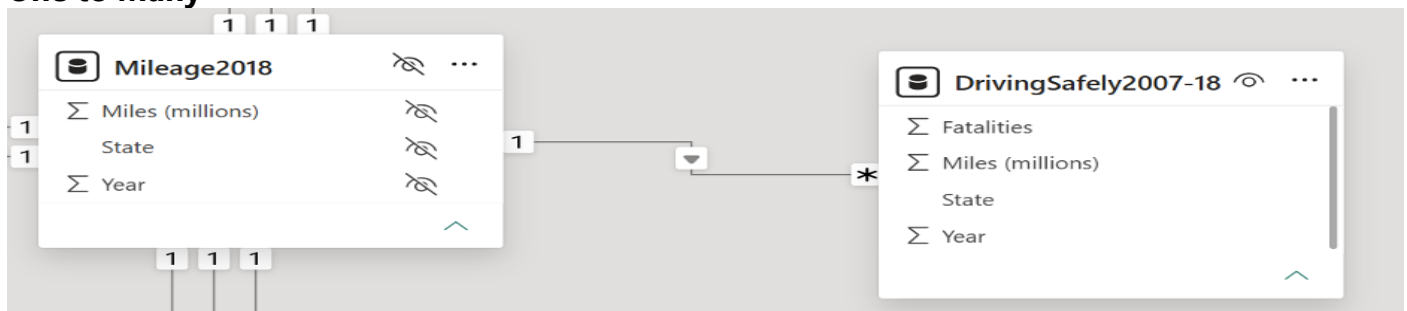
Without relationships, Power BI cannot correctly understand how tables are connected.

Cardinality and Relationship Types

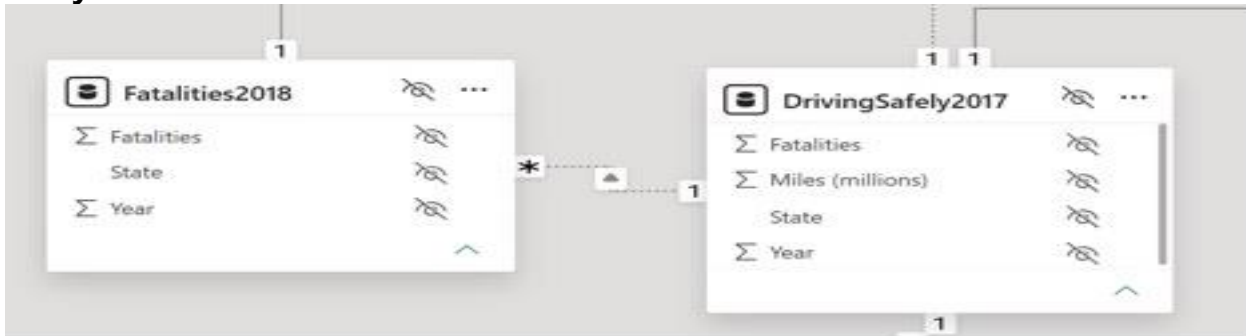
Cardinality describes how rows in one table relate to rows in another table. Power BI’s model view, cardinality is shown using the **1** and ***** icons at the ends of the relationship line.

Types	Definition
One-to-Many	One row in a dimension table relates to many rows in a fact table. This is the most common and preferred relationship type in Power BI models.
Many-to-One	Many rows in a fact table relate to one row in a dimension <u>table</u> . This is the reverse view of one-to-many.
One-to-One	One row in one table relates to one row in another <u>table</u> . This is less common in reporting models.
Many-to-Many	Many rows in one table relate to many rows in another table. This should be used carefully because it can create ambiguity and unexpected results.

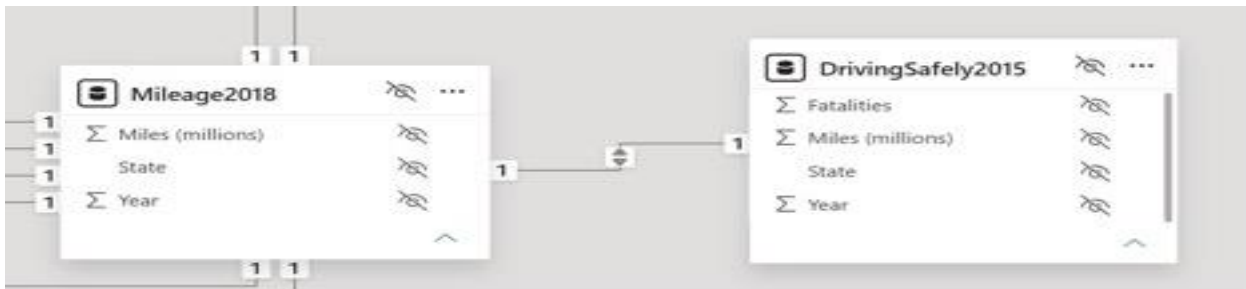
One to Many



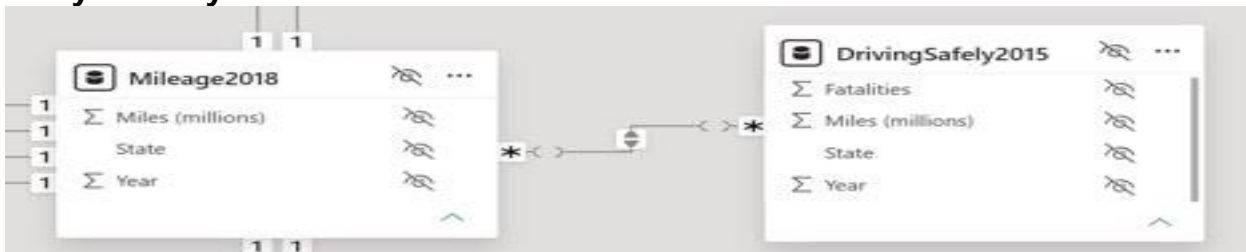
Many to One



One to One



Many to Many



Cross-Filter Direction

Cross-filter direction controls how filters move between related tables. In Power BI model view:

- Arrows show the filter direction
- A solid line shows an active relationship
- A dotted line shows an inactive relationship

Edit relationship

Select tables and columns that are related.

From table

DrivingSafely2007-18

Fatalities	Miles (millions)	State	Year
1110	61410	Alabama	2007
82	5153	Alaska	2007
1071	62963	Arizona	2007

To table

Mileage2018

Miles (millions)	State	Year
71167.20675	Alabama	2018
5487.319994	Alaska	2018
66144.51121	Arizona	2018

Cardinality

Many to one (*:1)

- ☒ Make this relationship active
- ☐ Assume referential integrity

Cross-filter direction

Single

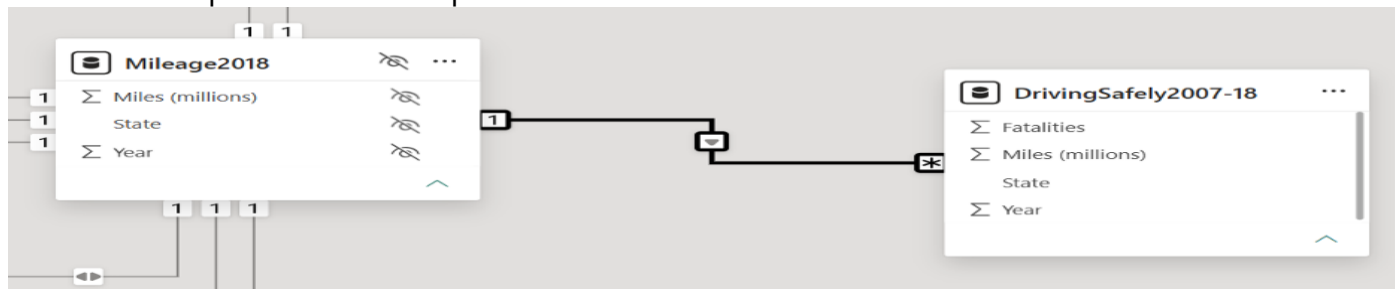
- ☒ Single
- ☐ Both

Save

Cancel

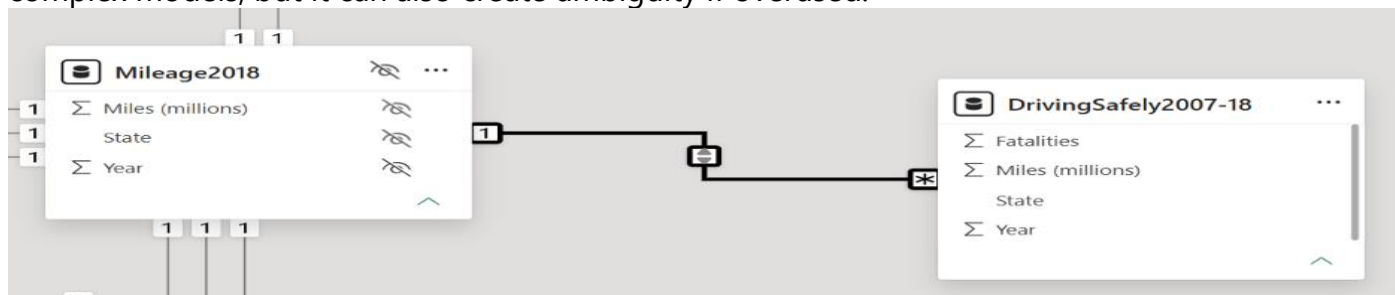
Single Direction

Filters usually flow from the dimension table to the fact table. This is often the preferred design because it keeps the model simpler and easier to understand



Both Direction

Filters can flow in both directions. This can be useful in some cases, such as bridging tables or more complex models, but it can also create ambiguity if overused.

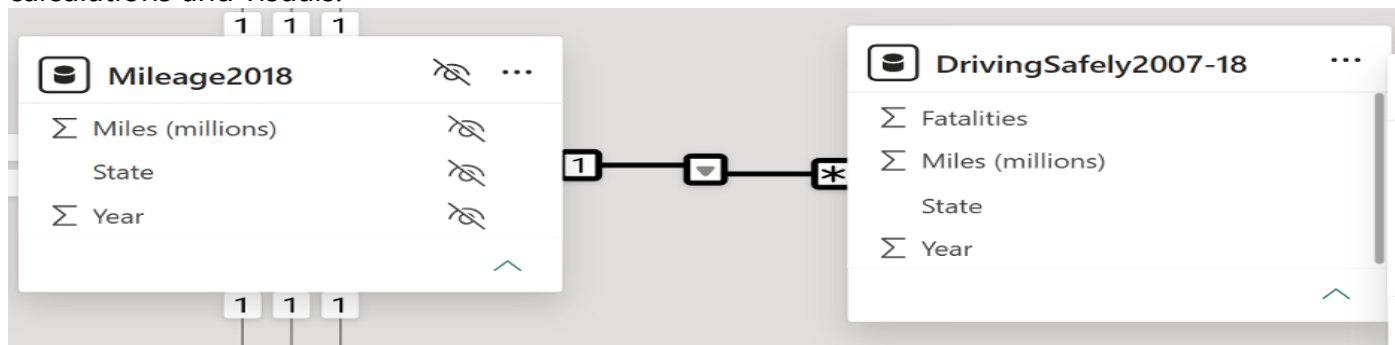


A good rule: Use single-direction filtering by default. Use both-direction filtering only when there is a clear modelling reason.

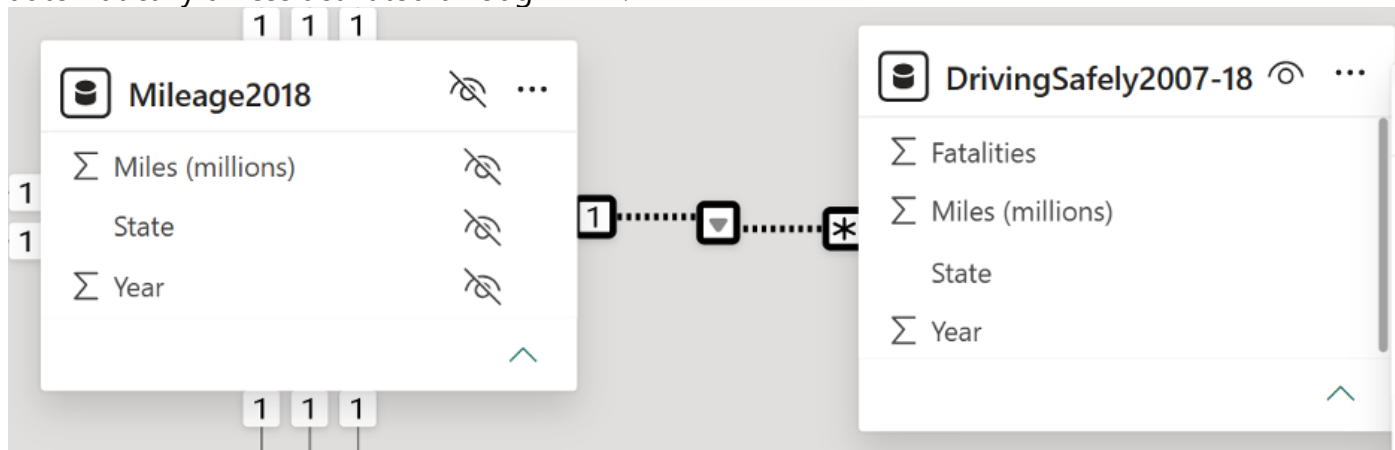
Active and Inactive Relationships

Power BI allows only one active relationship between two tables for a specific filter path.

Active Relationship: An active relationship is shown as a solid line and is used automatically in calculations and visuals.



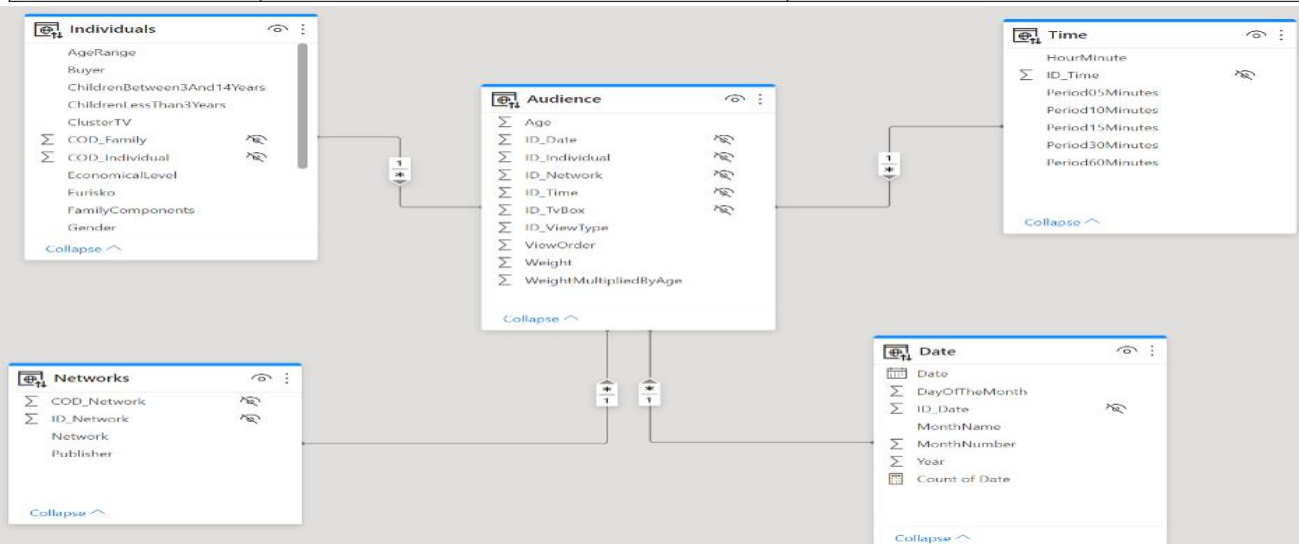
Inactive Relationship: An inactive relationship is shown as a dotted line and does not filter automatically unless activated through DAX.



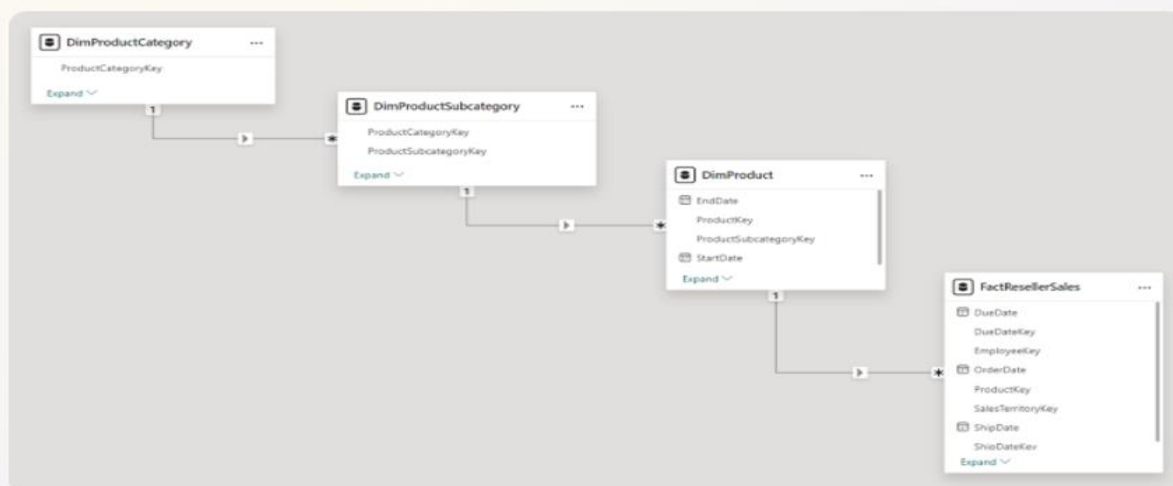
Schema Design: Star Schema and Snowflake Schema

Schema design describes how tables are arranged in the model. The two common patterns are star schema and snowflake schema.

Area	Star Schema	Snowflake Schema
Definition	A star schema has one or more fact tables in the <u>centre</u> , connected directly to dimension tables around them. A star schema is usually preferred in Power BI because it is simple, fast, and easier for report users to understand. The trade-off is that dimension tables may contain repeated descriptive information due to <u>denormalisation</u> .	A snowflake schema breaks dimension tables into smaller related tables. This reduces redundancy and can save storage, but it also creates more joins and a more complex model. Snowflake schemas may be useful in complex data environments, but they can be harder to design, maintain, and use for reporting.
Structure	Fact table connected directly to dimensions	Dimensions split into sub-dimensions
Design	<u>Denormalised</u>	<u>Normalised</u>
Ease of use	Easier for analysts and report users	More complex
Query performance	<u>Usually</u> faster	Can be slower due to more joins
Storage	More repeated data	Less repeated data
Best suited for	Power BI reporting and analysis	Complex or highly <u>normalised</u> environments



Snowflake example

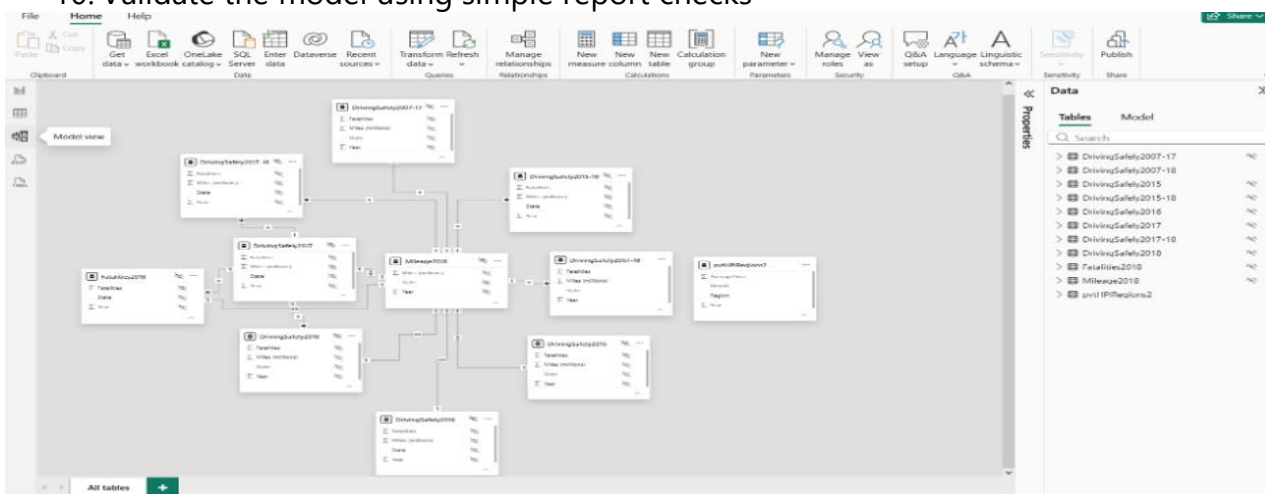


How to Configure Data Modelling in Power BI

Before loading data into the model, analysts should check Power BI Desktop's data load options. Power BI can automatically detect or import relationships from the source. These options can be useful, but analysts should still inspect and validate the final model carefully.

A simple modelling workflow:

1. Load cleaned tables into Power BI
2. Go to **Model view**
3. Review the tables and relationship lines
4. Open **Manage relationships**
5. Check relationship columns
6. Confirm cardinality
7. Confirm cross-filter direction
8. Check active and inactive relationships
9. Arrange the model visually for readability
10. Validate the model using simple report checks



Manage relationships

+ New relationship		⚡ Autodetect	✎ Edit	🗑 Delete	☰ Filter	▼
☐ From: table (column) ↑	Relationship	To: table (column)	Status			
☐ DrivingSafely2007-17 (State)	* — ◀ — 1	Mileage2018 (State)	Active			...
☐ DrivingSafely2007-18 (State)	* — ◀ — 1	Mileage2018 (State)	Active			...
☐ DrivingSafely2015 (State)	1 — ▶ — 1	Mileage2018 (State)	Active			...
☐ DrivingSafely2015-18 (State)	* — ◀ — 1	Mileage2018 (State)	Active			...
☐ DrivingSafely2016 (State)	1 — ▶ — 1	Mileage2018 (State)	Active			...
☐ DrivingSafely2017 (State)	1 — ▶ — 1	Mileage2018 (State)	Active			...
☐ ∑ DrivingSafely2017-18 (Fatalities)	* — ◀ — 1	∑ DrivingSafely2017 (Fatalities)	Inactive			...
☐ DrivingSafely2017-18 (State)	* — ◀ — 1	Mileage2018 (State)	Active			...
☐ ∑ DrivingSafely2018 (Fatalities)	* — ◀ — 1	∑ DrivingSafely2017 (Fatalities)	Inactive			...
☐ DrivingSafely2018 (State)	1 — ▶ — 1	Mileage2018 (State)	Active			...
☐ ∑ Fatalities2018 (Fatalities)	* — ◀ — 1	∑ DrivingSafely2017 (Fatalities)	Inactive			...
☐ Fatalities2018 (State)	◀ — ▶ — ◀	Mileage2018 (State)	Active			...

Close

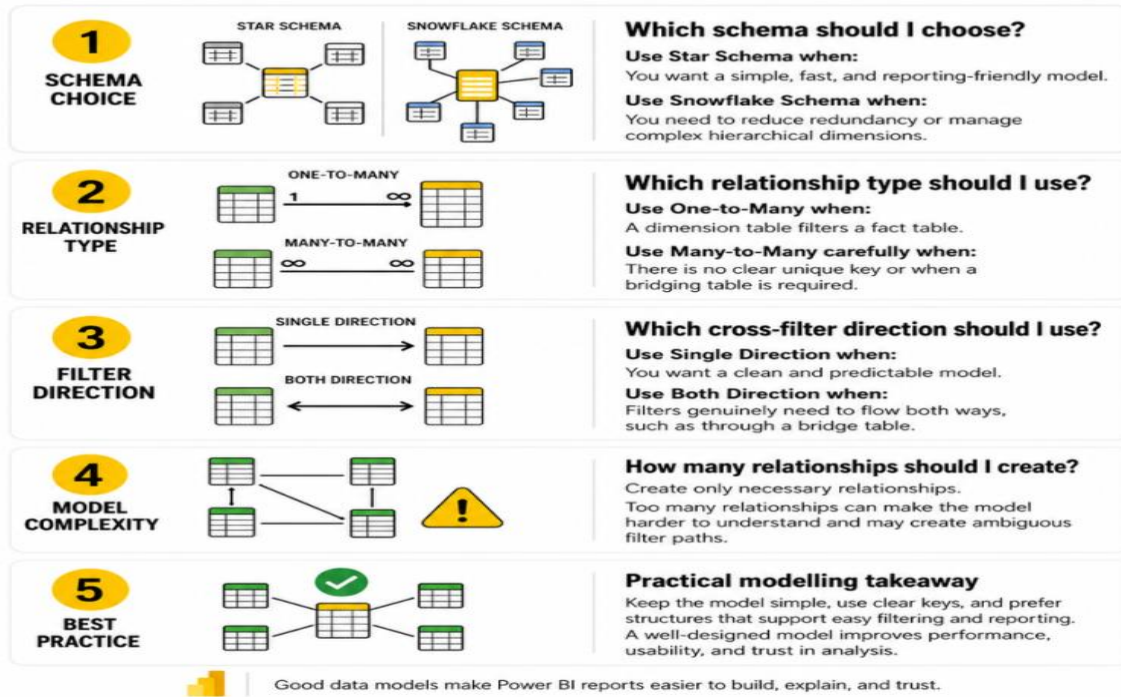
Close

Practical Way to Decide



A Simple Decision Guide for Power BI Data Modelling

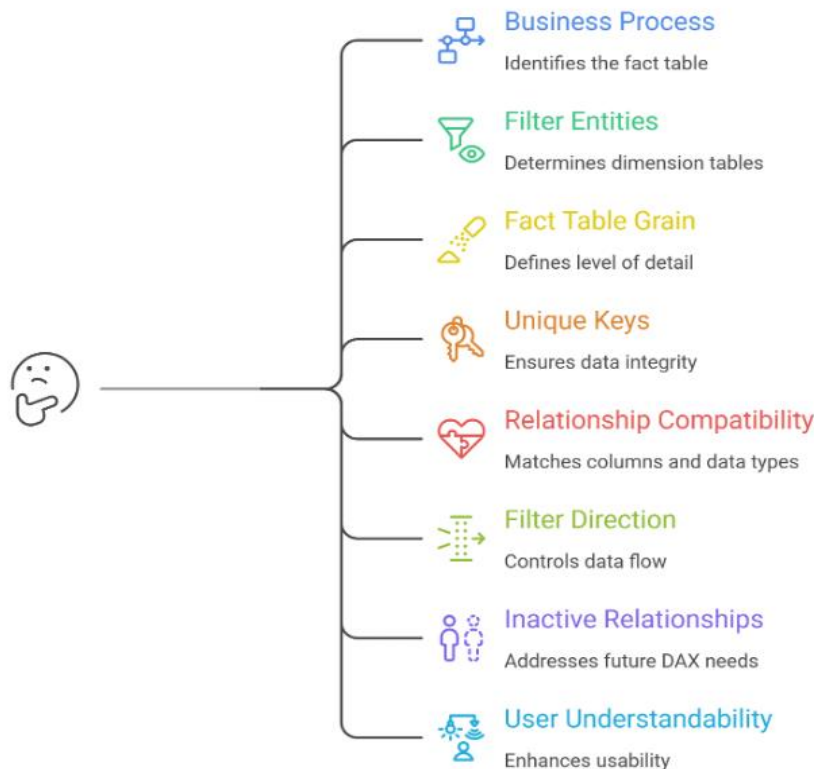
Choose the right modelling approach based on the design question you are solving.



Good data models make Power BI reports easier to build, explain, and trust.

What Analysts Should Ask Before Modelling

What questions should be asked before modelling?



Conclusion

Data modelling is where prepared data becomes structured for reliable analysis. Fact tables, dimension tables, relationships, cardinality, and schema design determine how Power BI filters data, calculates results, and supports business reporting.

A good model makes the report easier to build, easier to use, and easier to trust.

Stay tuned for next week's article in **The Insight Stack: Power BI, Layer by Layer**, where we move into **Week 6: DAX** and explore how analysts create reusable calculations and business logic inside Power BI.

Further Reading

- Data Modelling Overview: [Understand star schema and the importance for Power BI - Power BI | Microsoft Learn](#)
- One to One Relationship: [One-to-one relationship guidance - Power BI | Microsoft Learn](#)
- Many to Many Relationship: [Many-to-many relationship guidance - Power BI | Microsoft Learn](#)
- Reduce Data Model Size: [Data reduction techniques for Import modeling - Power BI | Microsoft Learn](#)